TASK 1: Logic Scoring of Preferences

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1 Description

For this task, the objective is to use the LSP method to create a decision support system for evaluating and selecting TV shows. The system considers five different factors, grouped into two main categories: Duration and Content. Each factor is scored based on the user's personal preferences, so utility functions need to be defined to reflect what the user values most. These functions help the system assess and recommend TV shows that match the user's tastes. The software used in this task was developed by the ITAKA research group at URV.

2 Utility Function Definition

This section outlines the different criteria and their corresponding utility functions, which define the user's preferences based on the values provided for each factor.

2.1 Episode Length

The episode length evaluates my preference regarding the Duration of each episode, which can range between 15 and 90 minutes. I define a utility function that reflects my preferred episode Duration within this range. Concretely, my preference relies on episodes that are long enough to take a break, meaning the Duration should be shorter than an hour, but long enough to avoid the need to watch more. This translates into low utilities for episodes shorter than half an hour, which increase up to a utility of 1 for episodes around 45 minutes, and decrease again as the episode length grows longer. This can be reflected in Figure 1.

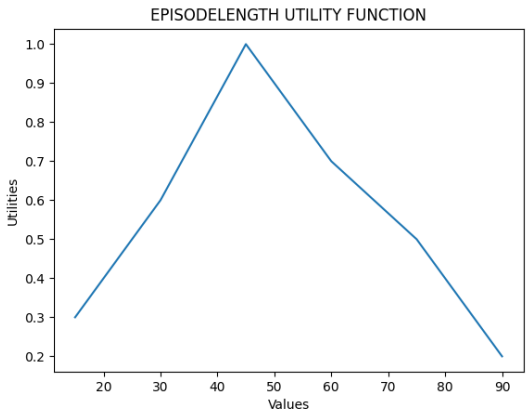


Figure 1: Utility function for Episode Length

2.2 Number of Episodes

This criterion evaluates my preference for the total number of episodes in the TV show, ranging from a minimum of 5 to a maximum of 80 episodes.

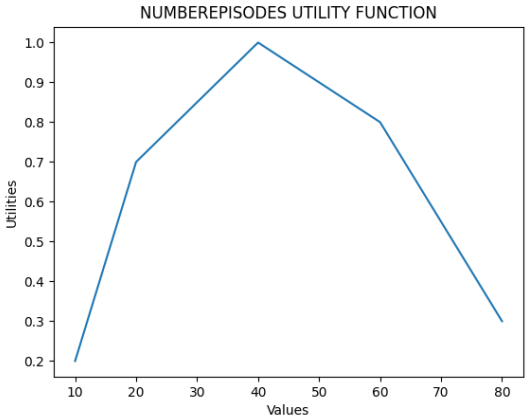


Figure 2: Utility function for Number of Episodes

I define a utility function to reflect my desired number of episodes within this range. As shown in Figure 2, I prefer shows that are neither too long nor too short. My most preferred TV shows are those that have 3 or 4 seasons, translating into a range of 30 to 50 episodes. This is where my utility function reach the highest values.

2.3 Action

This criterion is based on a score that indicates the level of action in the TV show. The action level is measured on a scale from 0 to 10, where 0 represents no action and 10 represents the highest level of action.

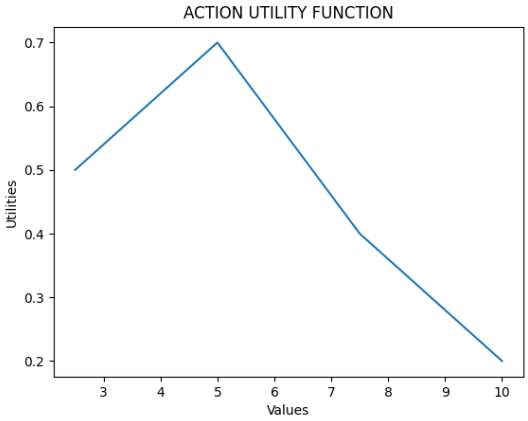


Figure 3: Utility function for Number of Action

Figure 3 shows my preferred level of action using this scale. I like shows with a bit of action, but not too much, as too much action can make it feel overwhelming. Therefore, my utility function increases at first but decreases significantly when the level of action gets too high. I prefer shows with good plots rather than those that focus too much on action.

2.4 Science Fiction

This criterion measures how much of the show is science fiction, using a score from 0 to 10. A score of 10 indicates the show is entirely science fiction, while a score of 0 indicates the show is completely based in reality.

I either like shows that are completely real, based on true stories, or shows that are entirely science fiction, while I do not enjoy shows that mix science fiction with reality. Therefore, my utility function depicted in Figure 4 is highest at the extremes: no science fiction or full science fiction. I prefer shows

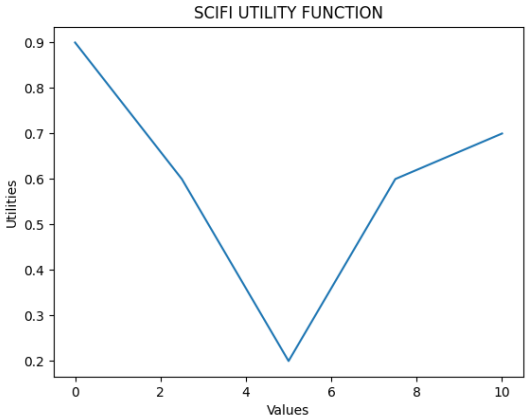


Figure 4: Utility function for Science Fiction

that are fully grounded in reality or entirely based on science fiction, and I assign slightly less utility to high sci-fi than high reality.

2.5 Comedy

This last criterion measures the percentage of comedy scenes in the show, with a score from 0 to 100. A score of 100 indicates the show is entirely comedy, while a score of 0 represents a completely non-comedic show, such as a drama.

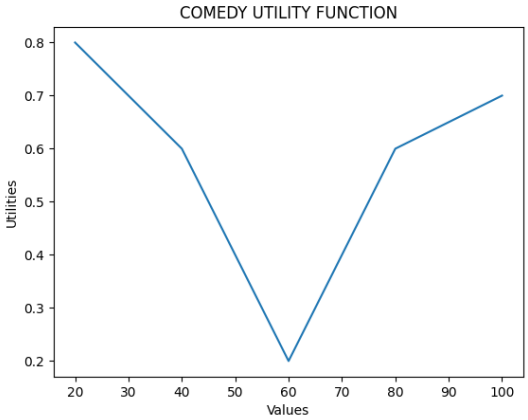


Figure 5: Utility function for Comedy

Figure 5 shows the utility function with a behavior similar to that of the science fiction criterion, where I prefer shows at the extremes, either fully comedic or fully dramatic. I do not enjoy shows that mix comedy and drama.

3 LSP Calculation

In this section, the TV show alternatives described in the statement will be analyzed by applying the utility functions to evaluate their suitability based on the defined criteria explained previously.

3.1 Suitability Results

This section shows and analyses the suitability results based on the preferences and utility functions defined for the TV shows, and evaluated on the alternatives presented in Table 1. From all the series in the statement, the result analysis will focus on the series Game Of Thrones, Big Bang and Bodies. As these are series that I have seen. Therefore, the evaluation will be more precise.

Name	Episode mins	Number of Episodes	Action	Science Fiction	Comedy
Fool me once	50	8	3	0	100
Big Bang	20	80	0	2	70
Bodies	65	8	7	4	0
Game of Thrones	75	50	9	5	0
Resilient Alien	44	12	4	4	80

Table 1: TV Show Alternatives

The utility rankings in Figure 6 align well with my theoretical preferences, particularly in ranking Game of Thrones as the best overall show (highest TVShowScore). This makes sense because it offers a consistent balance in terms of Content, Duration, and other factors that I value.

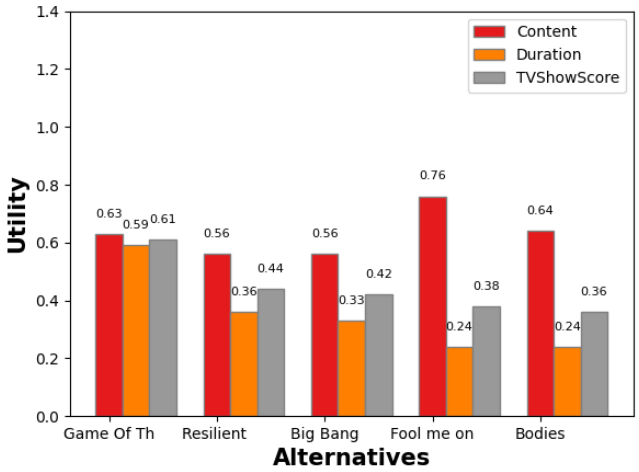


Figure 6: Suitability Plot

Regarding Bodies, I found it the least enjoyable, primarily due to its short length, which is reflected in its lower TVShowScore. However, its Content score is higher than Big Bang, which doesn't fully align with my experience. I actually prefer Big Bang's Content, but my general dislike for comedy influenced the rankings. Big Bang Theory is an exception for me, as its intelligent humor appeals to a tech and science audience, which I appreciate. Despite this, my usual preference against comedy caused Big Bang to rank lower than Bodies in Content. However, this did not impact the overall TVShowScore, as Big Bang still ranked higher than Bodies. Therefore, while the results are logical based on the system's metrics, they don't entirely reflect my personal preferences.

Performance Table

	EpisodeLength	NumberEpisodes	Action	Comedy	SciFi	Content	Duration	TVShowScore
Fool me once	0.9	0.2	0.54	0.7	0.9	0.76	0.24	0.38
Big Bang	0.4	0.3	0.5	0.4	0.66	0.56	0.33	0.42
Bodies	0.63	0.2	0.46	0.8	0.36	0.64	0.24	0.36
Game Of Thrones	0.5	0.9	0.28	0.8	0.2	0.63	0.59	0.61
Resilient Alien	0.97	0.3	0.62	0.6	0.36	0.56	0.36	0.44

Figure 7: Performance Table

On the other hand, Game of Thrones stands out as the most balanced and consistent across all evaluated criteria, aligning well with my personal experience and justifying its high ranking. In contrast, since I haven't watched Fool Me Once or Resilient Alien, I can't offer further insights into their rankings.

4 Polarity of the Aggregation Explanation

This section will focus on the explanation of the aggregation operators used in this system. These explanations are based on Figure 8 extracted from the class slides of the course PMCDSS at the URV.

Type of polarization	Level of polarization	Symbol	Orness	Andness
Disjunctive polarization	Strongest	D	1	0
	Very Strong	D++	0,93	0,06
	Strong	D+	0,87	0,12
	Medium Strong	D+-	0,81	0,18
	Medium	DA	0,75	0,25
	Medium Weak	D-+	0,68	0,31
	Weak	D-+	0,6	0,37
	Very weak	D--	0,56	0,43
Neutrality		A	0,5	0,5
Conjunctive polarization	Very weak	C--	0,43	0,56
	Weak	C-	0,37	0,6
	Medium Weak	C-+	0,31	0,68
	Medium	CA	0,25	0,75
	Medium Strong	C+-	0,18	0,81
	Strong	C+	0,12	0,87
	Very Strong	C++	0,06	0,93
	Strongest	C	0	1

Figure 8: LSP aggregation operators

4.1 TVShowScore (CA - Medium Conjunctive Aggregation)

The TVShowScore aggregation follows a medium conjunctive approach (Orness: 0.25 — Andness: 0.75). This means that both Content and Duration need to be sufficiently high for a show to achieve a good overall score. Since it is more conjunctive than neutral, a poor score in either Content or Duration will significantly lower the TVShowScore, as there is limited compensation between the two factors.

For example: Suppose a show has a Content score of 0.8 but a Duration score of 0.3. Since the aggregation is more conjunctive, the overall TVShowScore will be significantly impacted and may be around 0.45, rather than a simple average.

4.2 Duration (C+ - Strong Conjunctive Aggregation)

The Duration criterion follows a strong conjunctive approach (Orness: 0.12 — Andness: 0.87). This means that for a TV show to score well in Duration, both episode length and number of episodes must align with the preferred range. Since this criterion is highly conjunctive, if either of these factors deviates too much from the ideal, the Duration score will drop considerably, making it a strict requirement.

For example: If the ideal episode length is 50 minutes and the ideal number of episodes is 30, but a TV show has episodes of 60 minutes and only 8 episodes, the Duration score will likely be very low (e.g., 0.2-0.3), even though one of the factors (episode length) is somewhat reasonable because the other is considerably low (number of episodes).

4.3 Content (DA - Medium Disjunctive Aggregation)

The Content criterion follows a medium disjunctive approach (Orness: 0.75 — Andness: 0.25). This means that a TV show can still receive a high Content score if it excels in at least one of its subcategories (Action, Sci-Fi, or Comedy), even if the others are lower. Since it is more disjunctive, strong performance in just one Content-related aspect can balance out weaknesses in other subcategories, allowing for some flexibility.

For example: If a show has a Comedy score of 0 but a Sci-Fi score of 0.9 and an Action score of 0.6, it may still get an overall Content score of around 0.75, since high values in one or two categories compensate for low values in others.

5 Testing

In this section, different tests are evaluated on the LSP. First, two fictive alternatives are introduced to evaluate the model. Test 1 meets both Duration and Content preferences but has one weak elementary

criterion. Test 2 excels in Duration but performs poorly in Content, assessing its influence on the final ranking. Test 3 and 4 are about modifying the importance and the polarities to see the effects on the results.

5.1 Test 1: High Overall Satisfaction but Weakness in One Criterion

This first test adds the alternative *Test 1* to the system, which represents a show that satisfies the preferences on both Duration and Content, but with one elementary criterion with low satisfaction performance, in this case the Action. Having high utilities on all criteria apart from action results in a high overall score of TVShowScore of 0.88, as seen on row Test 1 of Figure 9.

LSP Method - iTAKA

File Help

Performance Table

	EpisodeLength	NumberEpisodes	Action	Comedy	SciFi	Content	Duration	TVShowScore
Fool me once	0.9	0.2	0.54	0.7	0.9	0.76	0.24	0.38
Big Bang	0.4	0.3	0.5	0.4	0.66	0.56	0.33	0.42
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Game of Thrones	0.5	0.9	0.28	0.8	0.2	0.63	0.59	0.61
Resilient Alien	0.97	0.3	0.62	0.6	0.36	0.56	0.36	0.44
Test 1	1	1	0.2	0.8	0.9	0.78	1.0	0.88
Test 2	1	1	0.2	0.2	0.36	0.29	1.0	0.47

Figure 9: Performance plot with new alternatives

5.2 Test 2: Good Duration but Poor Content

This second tests adds a show with good performance in the Duration but a poor performance in the Content criterion showed as *Test 2* in the Figure 9. As expected, the result shows a low TVShowScore of 0.47. Which is higher than that of the average because of the polarity aggregator.

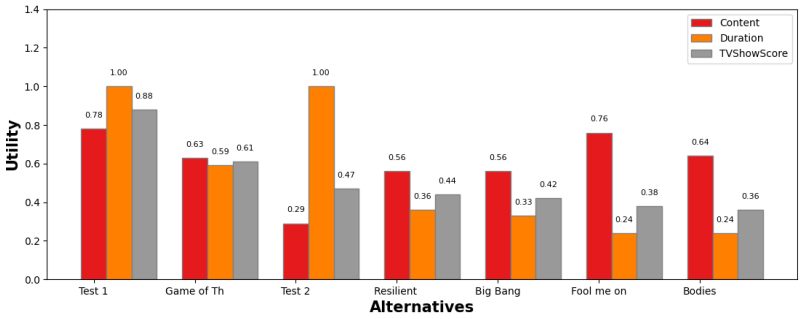


Figure 10: Suitability Plot with alternatives for Test 3 & 4

5.3 Test 3: Increased Importance of the Duration Criterion

In this test, the importance of the Duration criterion is increased from the original value of 0.5 to 0.75. Which means that the Content has been reduced from a weight of 0.5 to 0.25. The effect on the final ranking is observed in Figure 11.

By increasing the weight of Duration, we expect the shows with higher utility in the Duration category to be ranked higher, even if their performance in other criteria may be weaker.

We can evaluate the results by comparing Figure 10 with the Figure 11 that contains the new values. Concretely, what we can observe is that some alternatives have increased their overall TVShowScore and others have it decreased. We can conclude that the shows that have a higher Duration value than that of the Content, have increased its overall TVShowScore. That is, just the new two alternatives added, which are *Test 1 and Test2* while the other shows have a lower TVShowScore.

Performance Table

	EpisodeLength	NumberEpisodes	Action	Comedy	SciFi	Content	Duration	TVShowScore
Fool me once	0.9	0.2	0.54	0.7	0.9	0.76	0.24	0.3
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Game of Thrones	0.5	0.9	0.28	0.8	0.2	0.63	0.59	0.6
Resilient Alien	0.97	0.3	0.62	0.6	0.36	0.56	0.36	0.4
Test 1	1	1	0.2	0.8	0.9	0.78	1.0	0.94
Test 2	1	1	0.2	0.2	0.36	0.29	1.0	0.65

Figure 11: Suitability Plot increasing the importance of the Duration

5.4 Test 4: Change in the Polarity of the Duration Aggregator

In this test, we return to the case of equal weights for Duration and Content, but we change the polarity of the Duration aggregator, concretely, to the A aggregator, which according to Figure 8, is a neutral operator.

Performance Table

	EpisodeLength	NumberEpisodes	Action	Comedy	SciFi	Content	Duration	TVShowScore
Fool me once	0.9	0.2	0.54	0.7	0.9	0.76	0.55	0.64
Big Bang	0.4	0.3	0.5	0.4	0.66	0.56	0.35	0.43
Bodies	0.63	0.2	0.46	0.8	0.36	0.64	0.42	0.51
Game of Thrones	0.5	0.9	0.28	0.8	0.2	0.63	0.7	0.66
Resilient Alien	0.97	0.3	0.62	0.6	0.36	0.56	0.64	0.6
Test 1	1	1	0.2	0.8	0.9	0.78	1.0	0.88
Test 2	1	1	0.2	0.2	0.36	0.29	1.0	0.47

Figure 12: Performance table changing the polarity of the Duration aggregator to A

Changing the polarity operator for Duration from C+ to A results in a neutral polarization. This means that no polarization will be applied, and the results will reflect the average of its elementary criteria. As shown in Figure 12 the result of TVShowScore is the average of the values of the Content and the Duration.

On contrast, if the polarization were changed from conjunctive (C+) to disjunctive, the overall results would shift, favoring shows that excel in at least one criterion rather than requiring a balanced performance across all of them. This means that a show with a very high score in either Duration or Content could rank higher, even if it performs poorly in other areas.